ADVANCED COMPILER ENGINEERING

LLVM Kaleidoscope Compiler Architecture & Implementation

Comprehensive Deep-Dive into Token Rules, Quadruple/Triple IR, AST Design, Error Handling, Memory Segments, and Target Architecture

Repository: llvm-kaleidoscope

Backend: LLVM IRBuilder & Target Machine

Language: C++14 / Modern OOP

Focus: Frontend, IR Rules & Output Segments

CORE GOALS

Project Objectives

3 Core Pillars

01

1. Hand-Written Lexing & Operator-Precedence Parsing

02

2. Polymorphic AST Construction & Scoped Semantic Analysis

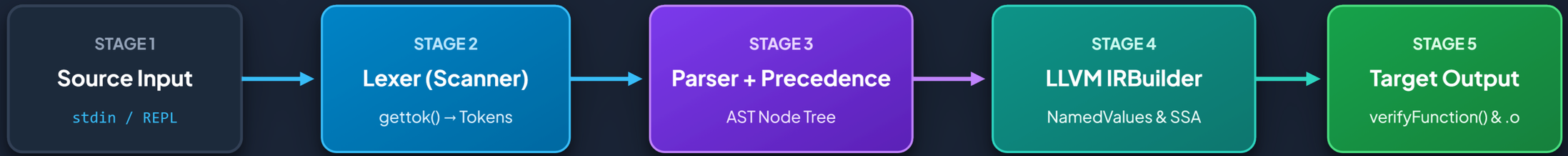
03

3. SSA LLVM IR Generation & Target Architecture Emission

SYSTEM ARCHITECTURE

High-Level Compiler Pipeline

End-to-End Pipeline



Interactive REPL Loop

Reads standard input continuously, classifies statements into definitions, externs, or immediate top-level expressions.

Verification & Segment Lowering

Emits validated LLVM byte-code into Module sections and dumps diagnostic text to standard output/error channels.

Complete Token Types & Lexer Rules

lexer/token.h & lexer.cpp

Complete Language Token Mapping

Token Category	Enum Identifier	Value	Lexer Match Pattern / Keyword
EOF / Stream	tok_eof	-1	End-of-File stream (EOF)
Function Defs	tok_def, tok_extern	-2, -3	Keywords "def", "extern"
Primary Entities	tok_identifier, tok_number	-4, -5	[a-zA-Z0-9]+, [0-9.]+ (→ double)
Control Flow	tok_if, tok_then, tok_else	-6, -7, -8	Keywords "if", "then", "else"
Loop Constructs	tok_for, tok_in	-9, -10	Keywords "for", "in"
Custom Operators	tok_binary, tok_unary	-11, -12	Keywords "binary", "unary"
Mutable Variables	tok_var	-13	Keyword "var"
Operators & Delims	ASCII direct	[0-255]	+ - * / < > = ( ) { } , ;

```
// Complete Token Definitions in lexer/token.h
enum Token {
  tok_eof = -1,
  // Commands & Definitions
  tok_def = -2, tok_extern = -3,
  // Primary Entities
  tok_identifier = -4, tok_number = -5,
  // Control Flow & Loops
  tok_if = -6, tok_then = -7, tok_else = -8,
  tok_for = -9, tok_in = -10,
  // User-Defined Operators & Variables
  tok_binary = -11, tok_unary = -12, tok_var = -13
};

// Lexical Scanner Match Dispatch in lexer/lexer.cpp
if (IdentifierStr == "def") return tok_def;
if (IdentifierStr == "extern") return tok_extern;
if (IdentifierStr == "if") return tok_if;
if (IdentifierStr == "then") return tok_then;
if (IdentifierStr == "else") return tok_else;
if (IdentifierStr == "for") return tok_for;
if (IdentifierStr == "in") return tok_in;
if (IdentifierStr == "var") return tok_var;
return tok_identifier;
```

Quadruple Rule in Intermediate Code Generation

4 Quadruple Structure: `(op, arg1, arg2, result)`

Our compiler strictly adopts the **Quadruple representation** where every intermediate operation explicitly specifies four components:

- **`op`:** The opcode / LLVM instruction type (e.g. `fadd`, `fmul`, `fcmp`, `call`).
- **`arg1`:** First input operand (variable, constant, or SSA register).
- **`arg2`:** Second input operand (or `null` for unary/return).
- **`result`:** Explicit named destination register (e.g. `%addtmp`, `%multmp`).

Why Quadruples in Modern LLVM?

Explicit SSA naming ensures compiler optimization passes (GVN, Dead Code Elimination) can safely reorder instructions without breaking statement index pointers.

Compiler Quadruple Instruction Set

Operation	Quadruple 4-Tuple	LLVM IRBuilder API Call
Addition	`(fadd, L, R, %addtmp)`	`Builder.CreateFAdd(L, R, "addtmp")`
Subtraction	`(fsub, L, R, %subtmp)`	`Builder.CreateFSub(L, R, "subtmp")`
Multiplication	`(fmul, L, R, %multmp)`	`Builder.CreateFMul(L, R, "multmp")`
Comparison	`(fcmp_ult, L, R, %cmptmp)`	`Builder.CreateFCmpULT(L, R, "cmptmp")`
Type Cast	`(uitofp, %cmptmp, double, %booltmp)`	`Builder.CreateUIToFP(L, Type, "booltmp")`
Function Call	`(call, @func, [Args], %calltmp)`	`Builder.CreateCall(CalleeF, ArgsV, "calltmp")`
Return	`(ret, %val, -, -)`	`Builder.CreateRet(RetVal)`

FRONTEND STAGE 2

Parser Architecture & Operator Precedence

parser/parser.cpp

Operator Precedence Climbing

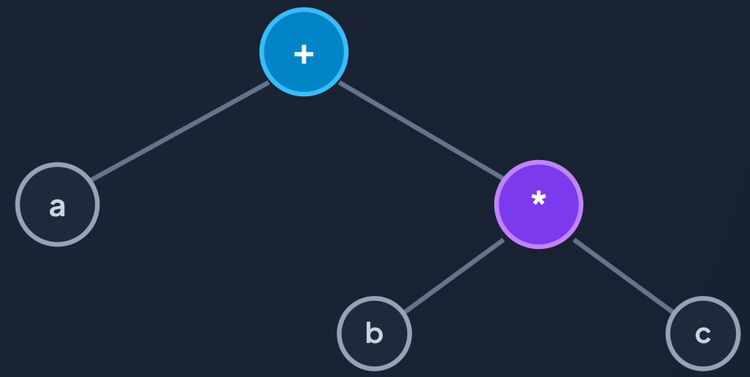
Uses recursive descent combined with operator precedence climbing for mathematical expressions.

Precedence Table

`<` : 10 | `+`, `-` : 20 | `*` : 40

- `ParsePrimary()` resolves numbers, variables, function calls, and parenthesized expressions.
- `ParseBinOpRHS()` dynamically groups subtrees based on comparative token precedence.

Expression Tree: `a + b * c`



Operator precedence ensures * binds tighter, forming the RHS subtree first.

INTERNAL REPRESENTATION

Abstract Syntax Tree (AST) Class Hierarchy

ast/ Directory

1 NumberExprAST

Numeric constants (e.g. `42.0`).

```
double Val;
codegen();
```

2 VariableExprAST

Variable reference (e.g. `x`).

```
string Name;
codegen();
```

3 BinaryExprAST

Binary operations (`LHS op RHS`).

```
char Op;
unique_ptr LHS, RHS;
```

4 CallExprAST

Function calls: `sin(x, y)`.

```
string Callee;
vector Args;
```

5 PrototypeAST

Captures function metadata: signature name and parameter identifiers list. Emits `llvm::Function*` declarations.

6 FunctionAST

Combines a `PrototypeAST` with a body `ExprAST`. Manages the function entry basic block and populates argument symbol tables.

Semantic Analysis: Complete Semantic Rules

Type Safety & Scoping Rules

R1 Scope & Binding

- **Formal Binding:** Arguments are registered into `NamedValues` map on function entry.
- **Undeclared Var Rule:** Referencing undeclared identifiers triggers `LogErrorV("Unknown variable name")`.
- **Scope Isolation:** `NamedValues.clear()` guarantees clean scopes across functions.

R2 Type System

- **Double Unification:** All numeric literals and expressions resolve to `getDoubleTy()`.
- **Boolean Promotion:** Comparison < produces `i1` and is coerced to `double` via `CreateUIToFP`.
- **Type Compatibility:** Math operands must both resolve to valid `llvm::Value*`.

R3 Function Arity

- **Existence Precondition:** Calls require callee to be declared in `TheModule`.
- **Strict Arity Rule:** Passed args must match signature: `CalleeF->arg_size() == Args.size()`.
- **Extern Linkage:** Prototypes create `ExternalLinkage` symbols.

R4 SSA & Control Flow

- **Condition Evaluation:** Conditions evaluate as truthy if $\neq 0.0$ via `CreateFCmpONE`.
- **Phi-Node Type Matching:** Both branch arms must have identical return types.
- **Terminator Invariant:** Basic blocks must end with `ret` or `br`.

Error Handling Based on All Token Rules

Token Type / Rule	Trigger Condition	Error Diagnostic Emitted	Compiler Recovery Action
`tok_identifier` (Prototype)	Missing function name after `def`/`extern`	`"Expected function name in prototype"`	Returns `nullptr` via `LogErrorP`
`'('` (Prototype Open)	Missing opening parenthesis in parameter list	`"Expected '(' in prototype"`	Returns `nullptr` via `LogErrorP`
`')'` (Prototype Close)	Missing closing parenthesis after parameters	`"Expected ')' in prototype"`	Returns `nullptr` via `LogErrorP`
`')'` (Paren Expression)	Unmatched closing parenthesis in `(expr)`	`"Expected )"`	Returns `nullptr` via `LogError`
`','` / `')'` (Call Args)	Missing comma or closing paren in argument list	`"Expected ')' or ',' in argument list"`	Aborts call node construction
`tok_identifier` (Scope)	Identifier not found in `NamedValues` symbol table	`"Unknown variable name"`	Returns `nullptr` via `LogErrorV`
Function Arity Rule	Argument count $\neq$ prototype parameter count	`"Incorrect # arguments passed"`	Aborts function call codegen
Codegen Failure Rule	Function body codegen returns `nullptr`	Transactional Module Rollback	`TheFunction->eraseFromParent()`

Output Channels & Target Memory Segments

Stream & Segment Mapping

1. Standard Output Streams

- **Standard Error (`stderr`):**
  - REPL interactive prompt: `ready>`
  - Diagnostics & Error Traps: `LogError: ...`
  - IR Emission feedback: `Read function definition:`
  - Final Module Dump: `TheModule->print(errs(), nullptr)`
- **Standard Input (`stdin`):**
  - Interactive command stream & file pipe execution (`./main < test.kld`)

2. Target Memory Segments

- **Text Segment (`.text` / Code):** Generated machine instructions for function basic blocks (e.g. `@foo`, `@__anon_expr`).
- **Read-Only Data (`.rodata` / Constant Pool):** Floating-point constants emitted via `ConstantFP::get(TheContext, APFloat(Val))`.
- **Symbol Table (`.symtab` / Relocations):** External declarations (`declare double @sin(double)`) and local symbol definitions.
- **Heap & Memory Arena:** AST ownership nodes managed via `std::unique_ptr` and LLVM IRContext allocation pools.

HARDWARE TARGET

Target Architecture & Native Binary Emission

LLVM TargetMachine & Object Files

🎯 Target Triples

Defines cross-platform target hardware configurations:

```
"x86_64-pc-linux-gnu"
"x86_64-w64-windows-msvc"
"aarch64-apple-darwin"
```

- `InitializeAllTargetInfos()`
- `InitializeAllTargets()`
- `InitializeAllAsmPrinters()`

⚙ TargetMachine & ABI

Configures CPU capabilities, ABI conventions, and memory alignments:

- **DataLayout:** Dictates 64-bit pointer size and memory alignments.
- **CPU Architecture:** Targets specific CPUs (e.g. `generic`, `haswell`).
- **Relocation Model:** Configured for Position Independent Code (`Reloc::PIC_`).

📦 Object File Emission

Lowers LLVM IR directly into binary object files (`.o` / `.obj`):

```
legacy::PassManager pass;
auto FileType = CGFT_ObjectFile;
TheTargetMachine→
  addPassesToEmitFile(pass, dest, nullptr, FileType);
pass.run(*TheModule);
```

Linkable via `clang/gcc/lld` into native executables.

CONCRETE TRACE & OUTPUT

Execution Trace & Output Segment Breakdown

def foo(x y) x + y * 4.0;

1. Tokens & Grammar

```
[tok_def] "def"
[tok_identifier] "foo"
['(']
[tok_identifier] "x"
[tok_identifier] "y"
[')']
[tok_identifier] "x"
['+']
[tok_identifier] "y"
['*']
[tok_number] 4.0
[';']
```

2. Quadruple / AST Lowering

```
// Quadruple Form
(0) (FMUL, y, 4.0, %multmp)
(1) (FADD, x, %multmp, %addtmp)
(2) (RET, %addtmp, -, -)

// Scoping Check
NamedValues["x"] = %x
NamedValues["y"] = %y
```

3. Emitted Output & Segment

```
// Channel: stderr
Read function definition:
// Segment: .text (Code)
define double @foo(double %x, double %y) {
entry:
  %multmp = fmul double %y, 4.000000e+00
  %addtmp = fadd double %x, %multmp
  ret double %addtmp
}
```

ENGINEERING ROADMAP

Future Work Plan & Implementation Milestones

Work Plan Diagram

P1 Phase 1: JIT & Optimization

Milestone 1 • Chapter 4

- Integrate **LLVM OrcJIT / LLJIT** to evaluate top-level expressions immediately.
- Attach `FunctionPassManager` with GVN and Instruction Combining passes.

P2 Phase 2: Control Flow & State

Milestone 2 • Chapters 5–7

- Implement `IfExprAST` with conditional phi-nodes.
- Add `ForExprAST` loop construct.
- Add mutable local variables with stack `alloca` and LLVM `mem2reg`.

P3 Phase 3: Target Object Emission

Milestone 3 • Chapters 8–9

- Configure target machine triple and data layouts.
- Emit native object code (`.o` / `.obj`) and link to standalone executables.
- Add DWARF / CodeView debug symbols.

🎉 SUMMARY & WRAP-UP

Thank You! Questions & Discussion

Key Takeaways:

- Clear separation of concerns between Tokenizer, Grammar Parser, AST, and LLVM Backend.
- Comprehensive token-rule-based error handling ensuring graceful REPL recovery and transactional IR rollbacks.
- Production-grade lowering to verified SSA Quadruple IR and clear target architecture mapping for native binary compilation.

📖 **Reference:** llvm.org/docs/tutorial

📁 **Repository:** llvm-kaleidoscope